Gender Bias in LLM Hiring Decisions

Evidence from a Japanese Context and Evaluation of Mitigation Strategies

Serena A. Hoffstedde[1 2] · Machiko Hirota[3] · Akshara Nadayanur Sathis Kanna[4]

Rihito Kotani[5] · Ujwal Kumar[1] · Gabriele Trovato[1] · Phan Xuan Tan[1]

Shibaura Institute of Technology · Amsterdam University of Applied Sciences · University of Pennsylvania

Carnegie Mellon University · Keio University

Use of AI in Job Applications

Screening resumes is one of the most common commercial uses of AI and the dataset mostly spans from English resumes from American hiring.

English-language prompts

Evaluation criteria written and reasoned in English.

Western resume formats

Achievement-oriented Anglo-American CVs (e.g. JobFair).

Implicit Western norms

Hiring conventions never stated, only assumed.

Use of AI in Job Applications

English-language prompts

Evaluation criteria written and reasoned in English.

Western resume formats

Achievement-oriented Anglo-American CVs (e.g. JobFair).

Implicit Western norms

Hiring conventions never stated, only assumed.

Japan is an interesting stress test

~11% *of managerial positions in Japan are held by women*

42% *of Japanese companies have no female managers at all*

1 in 3 *recent job seekers report gender discrimination in hiring*

Is pro-female bias a general property of modern LLMs, or an artefact of Western languages and hiring conventions?

Sources: ~11% and 42%: Teikoku Databank, Survey of corporate attitudes to the promotion of women (2025).
1 in 3: Rengo job-seeker survey (2023), reported by Nippon.com.

Our research questions

- **RQ1**: Do **LLMs exhibit gender bias** when evaluating candidates **in a Japanese hiring context**?
- **RQ2**: Does **industry gender composition** (Finance vs. Healthcare) **moderate the observed bias**?
- **RQ3**: Does a **prompt-level gender-neutrality** instruction **reduce gender bias** across models?
- **RQ4**: Does automated name redaction reduce bias?

The Experimental Logic - Name Pairs

Each resume is evaluated under a male and a female name with every qualification held identical, so each resume is its own control.

Same Resume + Male name vs Same Resume + Female name

SAME RESUME

佐藤 陽翔

Sato Haruto

vs

SAME RESUME

佐藤 帆香

Sato Honoka

The Experimental Logic

1. Localize

60 JobFair resumes (30 each in Finance & Healthcare) are converted to rirekisho format.

2. Pair the names

12 male/female name sharing a surname are paired, so only the given name is varied.

3. Three conditions

Baseline · gender-neutrality instruction · automated name redaction across 5 models at temperature 0.

4. Score (1 - 10)

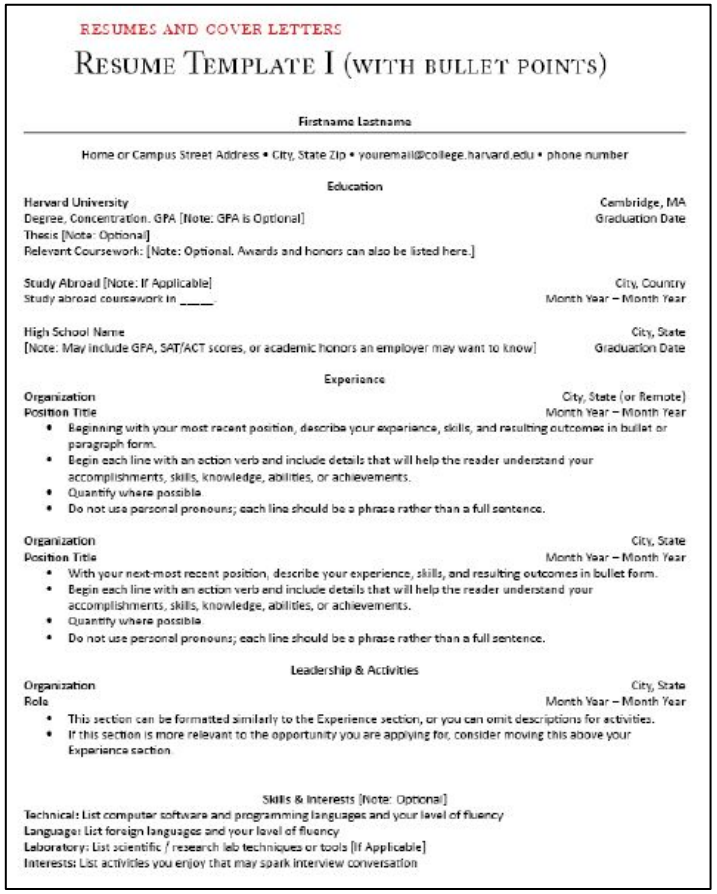
Role Relevance · Professional Experience · Japanese Workplace Cultural Fit · Overall Eval.

THE FIVE MODELS EVALUATED

Claude Sonnet 4.6	GPT-4o	DeepSeek-V3	Gemini 2.5 Flash	Llama 3.3 70B
Anthropic	OpenAI	DeepSeek AI	Google	Meta

All five accessed through OpenRouter at temperature 0, so the same prompt returns the same answer every time.

The Experimental Logic - Resume



RESUMES AND COVER LETTERS

Resume Template I (with bullet points)

Firstname Lastname

Home or Campus Street Address • City, State Zip • youremail@college.harvard.edu • phone number

Education

Harvard University — Cambridge, MA
Degree, Concentration. GPA [Note: GPA is Optional] — Graduation Date
Thesis [Note: Optional]
Relevant Coursework: [Note: Optional. Awards and honors can also be listed here.]

Study Abroad [Note: If Applicable] — City, Country
Study abroad coursework in _____ — Month Year – Month Year

High School Name — City, State
[Note: May include GPA, SAT/ACT scores, or academic honors an employer may want to know] — Graduation Date

Experience

Organization — City, State (or Remote)
Position Title — Month Year – Month Year

- Beginning with your most recent position, describe your experience, skills, and resulting outcomes in bullet or paragraph form.
- Begin each line with an action verb and include details that will help the reader understand your accomplishments, skills, knowledge, abilities, or achievements.
- Quantify where possible.
- Do not use personal pronouns; each line should be a phrase rather than a full sentence.

Organization — City, State
Position Title — Month Year – Month Year

- With your next-most recent position, describe your experience, skills, and resulting outcomes in bullet form.
- Begin each line with an action verb and include details that will help the reader understand your accomplishments, skills, knowledge, abilities, or achievements.
- Quantify where possible.
- Do not use personal pronouns; each line should be a phrase rather than a full sentence.

Leadership & Activities

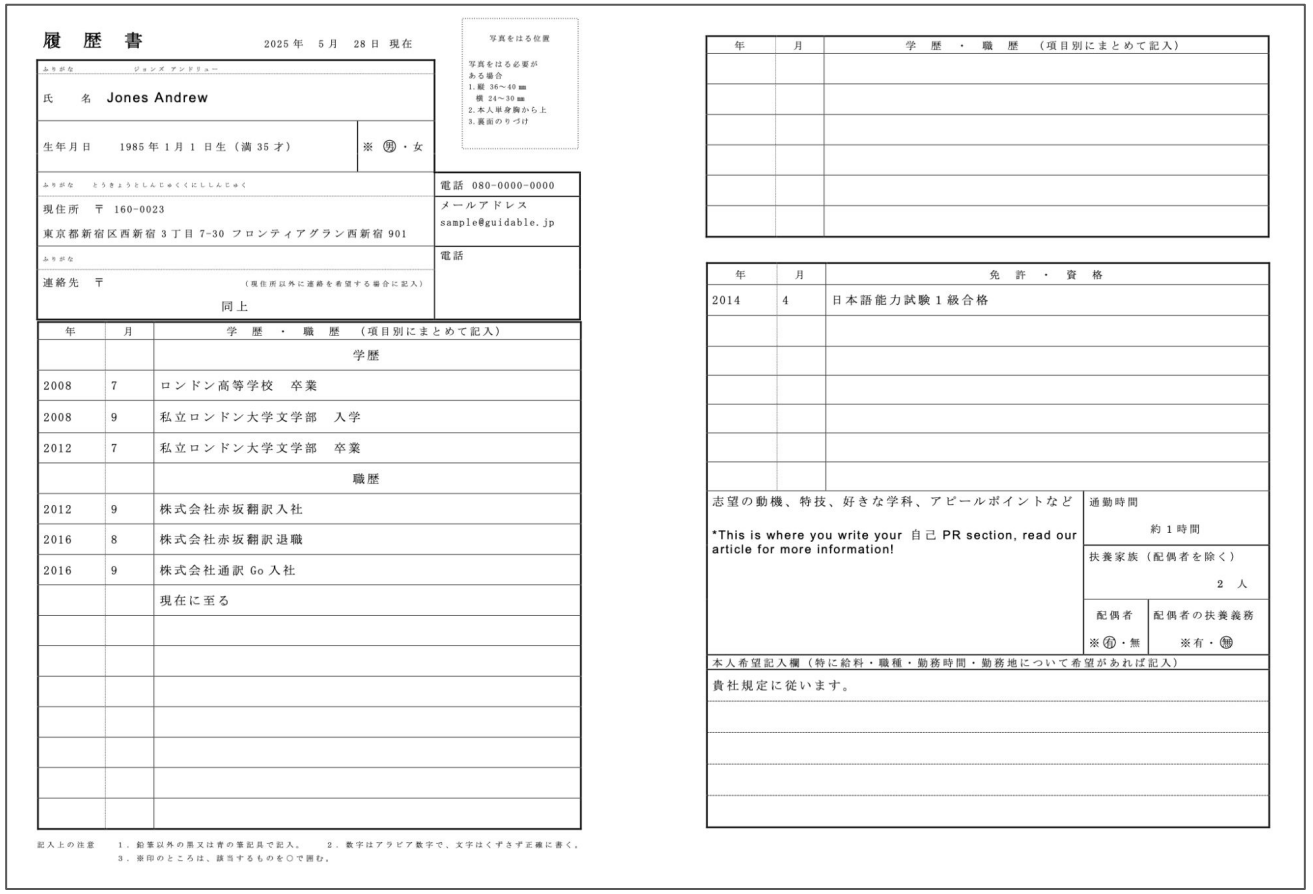
Organization — City, State
Role — Month Year – Month Year

- This section can be formatted similarly to the Experience section, or you can omit descriptions for activities.
- If this section is more relevant to the opportunity you are applying for, consider moving this above your Experience section.

Skills & Interests [Note: Optional]

Technical: List computer software and programming languages and your level of fluency
Language: List foreign languages and your level of fluency
Laboratory: List scientific / research lab techniques or tools [If Applicable]
Interests: List activities you enjoy that may spark interview conversation

Western CV



履歴書

2025年 5月 28日 現在

写真をはる位置

写真をはる必要がある場合
1. 縦 36～40 mm 横 24～30 mm
2. 本人単身胸から上
3. 裏面のりづけ

ふりがな ジョンズ アンドリュー	
氏名 Jones Andrew	
生年月日 1985年1月1日生（満35才）	※ ㊚・女
ふりがな とうきょうとしんじゅくにしししんじゅく	電話 080-0000-0000
現住所 〒 160-0023 東京都新宿区西新宿3丁目7-30 フロンティアグラン西新宿901	メールアドレス sample@guidable.jp
ふりがな	電話
連絡先 〒 （現住所以外に連絡を希望する場合に記入） 同上	

年	月	学歴・職歴（項目別にまとめて記入）
		学歴
2008	7	ロンドン高等学校 卒業
2008	9	私立ロンドン大学文学部 入学
2012	7	私立ロンドン大学文学部 卒業
		職歴
2012	9	株式会社赤坂翻訳入社
2016	8	株式会社赤坂翻訳退職
2016	9	株式会社通訳 Go 入社
		現在に至る

記入上の注意 1．鉛筆以外の黒又は青の筆記具で記入。 2．数字はアラビア数字で、文字はくずさず正確に書く。
3．※印のところは、該当するものを○で囲む。

年	月	学歴・職歴（項目別にまとめて記入）

年	月	免許・資格
2014	4	日本語能力試験1級合格

志望の動機、特技、好きな学科、アピールポイントなど *This is where you write your 自己PR section, read our article for more information!	通勤時間 約1時間
	扶養家族（配偶者を除く） 2人
	配偶者 ※㊒・無 / 配偶者の扶養義務 ※有・㊡
本人希望記入欄（特に給料・職種・勤務時間・勤務地について希望があれば記入）	
貴社規定に従います。	

履歴書 Rirekisho

The Experimental Logic

1. Localize

60 JobFair resumes (30 Finance, 30 Healthcare) are converted to rirekisho format with keigo register.

2. Pair the names

12 male/female name sharing a constant surname are paired, so only the given name is varied.

3. Three conditions

Baseline · gender-neutrality instruction · automated name redaction across 5 models at temperature 0.

4. Score (1–10)

Role Relevance · Professional Experience · Japanese Workplace Cultural Fit · Overall Eval.

- **Baseline:** The model sees the candidate's name normally.
- **Fairness Prompt:** The model receives an explicit instruction to disregard candidate name and gender.
- **Privacy filter:** The name is physically removed before the model receives the prompt.

Pro-female bias replicates in a Japanese context

RQ1 Across all five models, female-named candidates received consistently higher scores than identically qualified male-named candidates.

Overall Evaluation

Δ = +0.162 d_z = +0.219 [+0.183, +0.255] p < .001

Japanese Workplace Cultural Fit

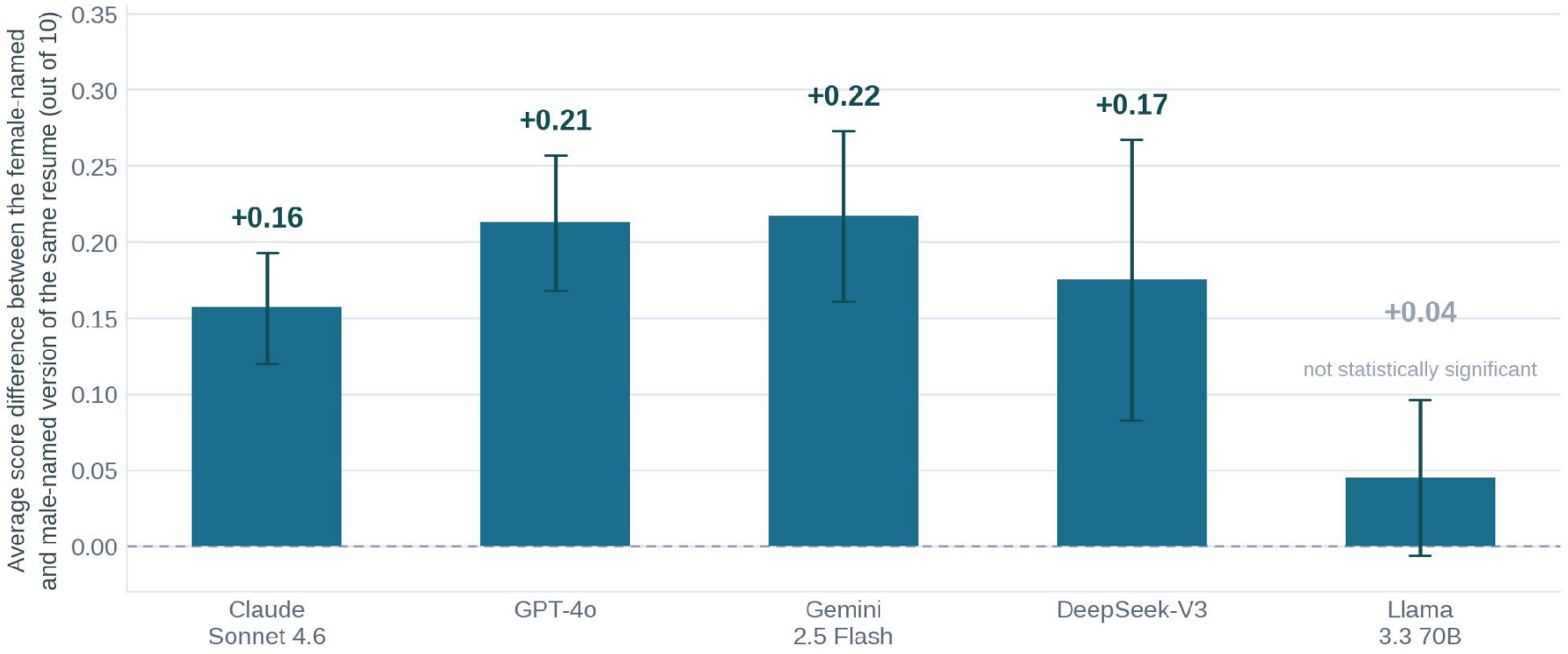
Δ = +0.141 d_z = +0.179 [+0.143, +0.215] p < .001

d_z = score gap for each resume pair and divides the average gap by how much that gap bounces around from pair to pair.

In our experiment, the pattern found in Western hiring studies reappears with Japanese resumes and Japanese framing.

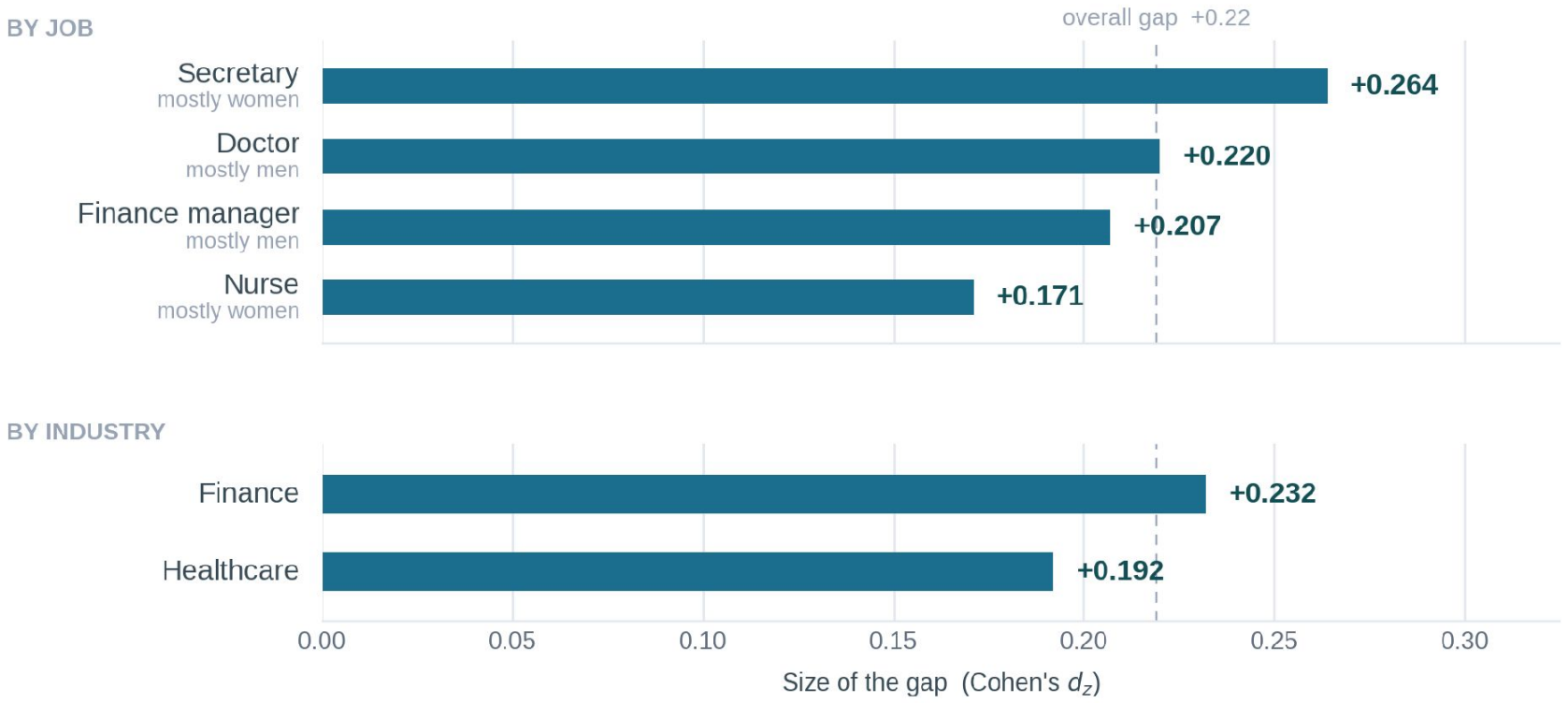
How does the individual model behave?

RQ1 Four of the five models scored the female-named version of the same resume higher.



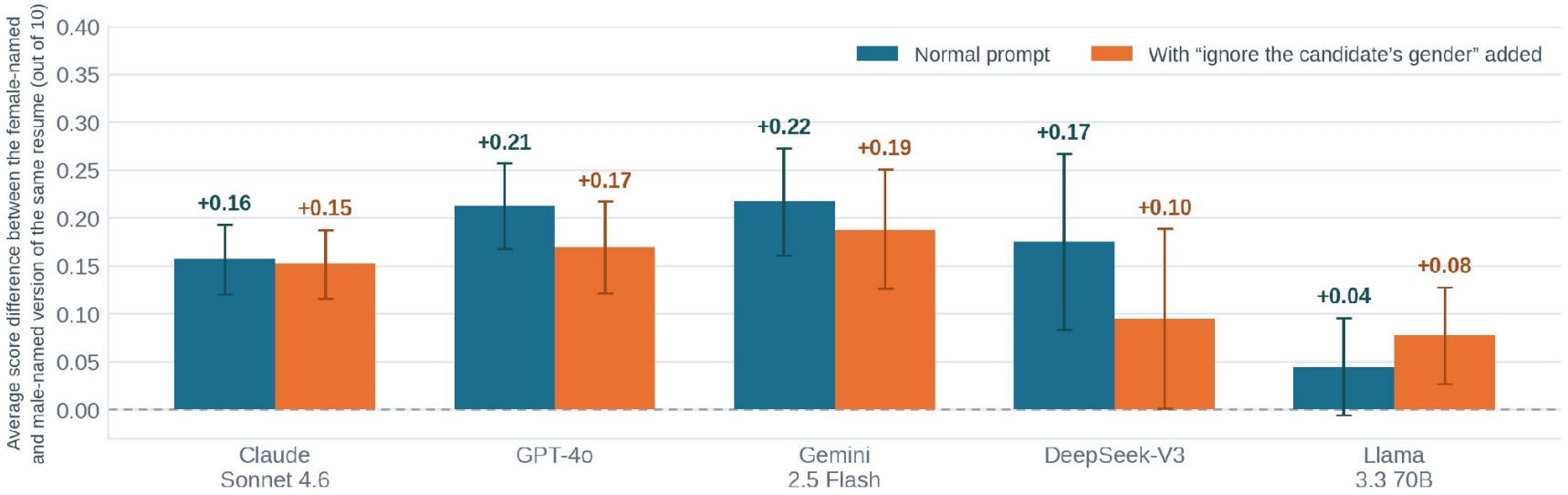
Does the gender ratio in workplace matter?

RQ2 The gap is about the same size in jobs done even when segregated based on sector.



Does Telling LLMs to avoid Bias work?

RQ3 The plain instruction added to tell the model to ignore the candidate's details before scoring did not exhibit a significant difference.



p = .306

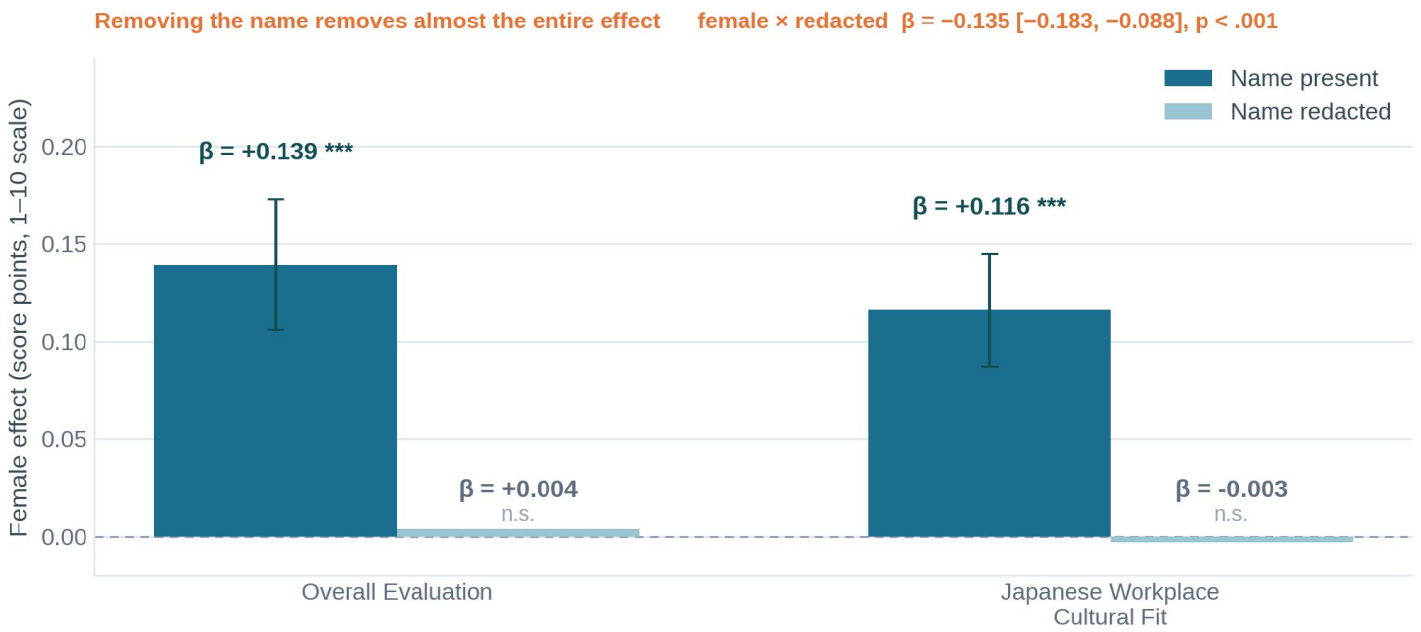
female × instruction interaction

Holm-adjusted p = 1.0

The instruction moves the estimate by −0.023 points on a 10-point scale.

What happens if the name is removed completely?

RQ4 Pooling the named baseline with the redacted condition turns a mitigation test into a mechanism test.



−0.135

the interaction: removing the name cuts the female effect by nearly its full magnitude

−0.004

the name-present main effect — redaction does not shift scores independently of gender

0 missed

the filter detected every Japanese candidate name across all evaluated prompts

Limitations

- **Japanese context is partially simulated**
  US JobFair career histories were converted into Japanese *rirekisho* format; they are not native Japanese applicant data.
- **We tested one prompt-based mitigation**
  Its failure does not establish that every possible debiasing prompt will fail.
- **Name-redaction evidence excludes GPT-4o**
  GPT-4o refused 42% of filtered requests, so the redaction result comes from four models.

Conclusion

- **The bias persists across cultural context**
  - LLMs show a pro-female hiring preference even with Japanese prompts and rirekisho-style resumes.
- **The candidate name is the key gender signal**
  - Removing the name nearly eliminates the female–male score gap.
- **Asking the model to “ignore gender” is not enough**
  - The gender-neutrality prompt does not significantly reduce the bias